Course Title: Ethical Hacking

Objective

The objective of this experiment is to perform reconnaissance and system monitoring using Sysmon, theHarvester, WHOIS, and Netdiscover. These tools help collect information about systems, domains, and networks, monitor system activities, identify active devices, and gather publicly available information for cybersecurity analysis and incident investigation.

Methodology

1. Install and configure Sysmon to monitor system activities and generate security event logs.
2. Use theHarvester to gather publicly available information such as email addresses, subdomains, hostnames, and IP addresses related to the target domain.
3. Perform WHOIS queries to retrieve domain registration details, registrar information, nameservers, and registration dates.
4. Use Netdiscover to scan the local network and identify active hosts, IP addresses, MAC addresses, and device vendors.
5. Analyze the information collected from all tools to understand the system, network, and domain environment and identify potential security risks.

Expected Outcome

By completing this experiment, you will be able to:

- Monitor system activities and generate detailed security logs using Sysmon.
- Collect publicly available information about target domains using theHarvester.
- Retrieve domain registration and ownership details using WHOIS.
- Discover active devices on a local network using Netdiscover.
- Gain practical experience with reconnaissance and system monitoring techniques.
- Improve the ability to analyze systems and networks for cybersecurity monitoring, incident response, and security assessments.

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1.Introduction to Sysmon (System Monitor)

Sysmon (System Monitor) is a free Windows system monitoring tool developed by Microsoft. It runs as a Windows service and records detailed system activity in the Windows Event Log, helping administrators and cybersecurity professionals detect suspicious or malicious behavior.

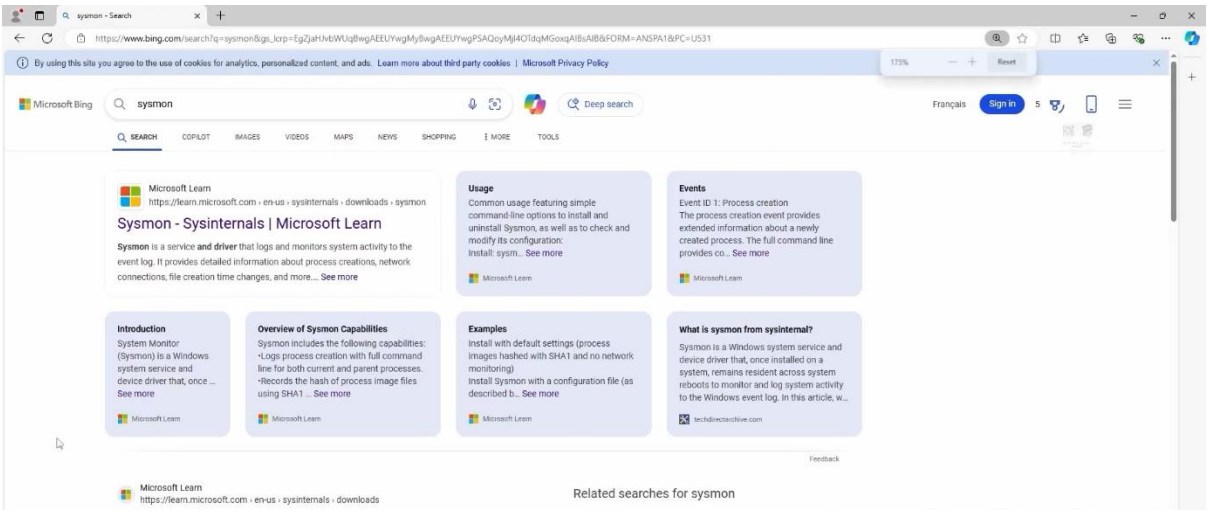
1.1. Uses of Sysmon

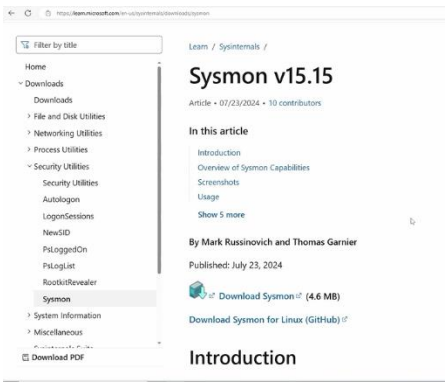
- Monitors process creation and termination.
- Records file creation time changes.
- Monitors network connections.
- Detects driver and DLL loading.
- Records registry modifications.
- Monitors file creation and deletion

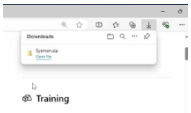
2. Setting up Sysmon

Step 1: Download Sysmon

Search for Sysmon and download Sysmon from the Sysinternals Suite.

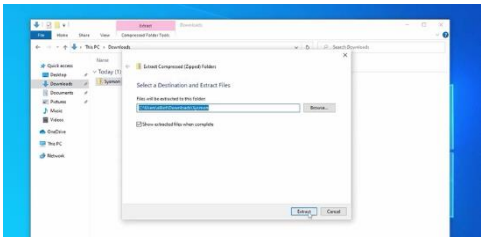






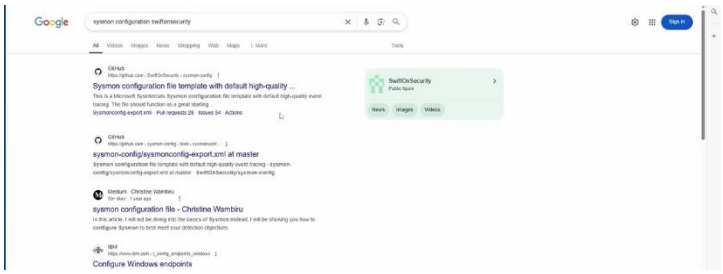
Step 2: Extract the Zip file

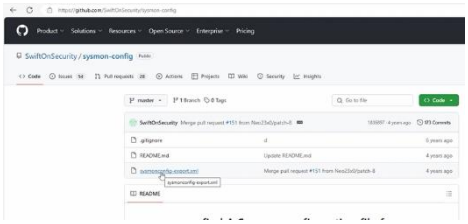
Extract the ZIP file from the downloaded file.



Step 3: Download configuration file

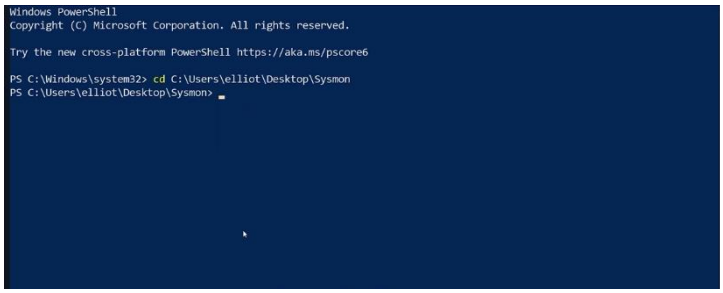
This installs Sysmon with custom monitoring rules defined in sysmonconfig.xml.

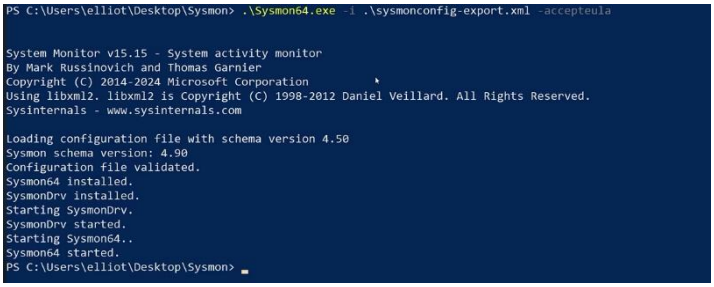




Step 4: Open Command Prompt

Open Command Prompt or PowerShell as Administrator.





Step 5: Install Sysmon

sysmon.exe -i

This installs the Sysmon service using the default configuration.



3. Sysmon Commands

3.1. Update the Configuration

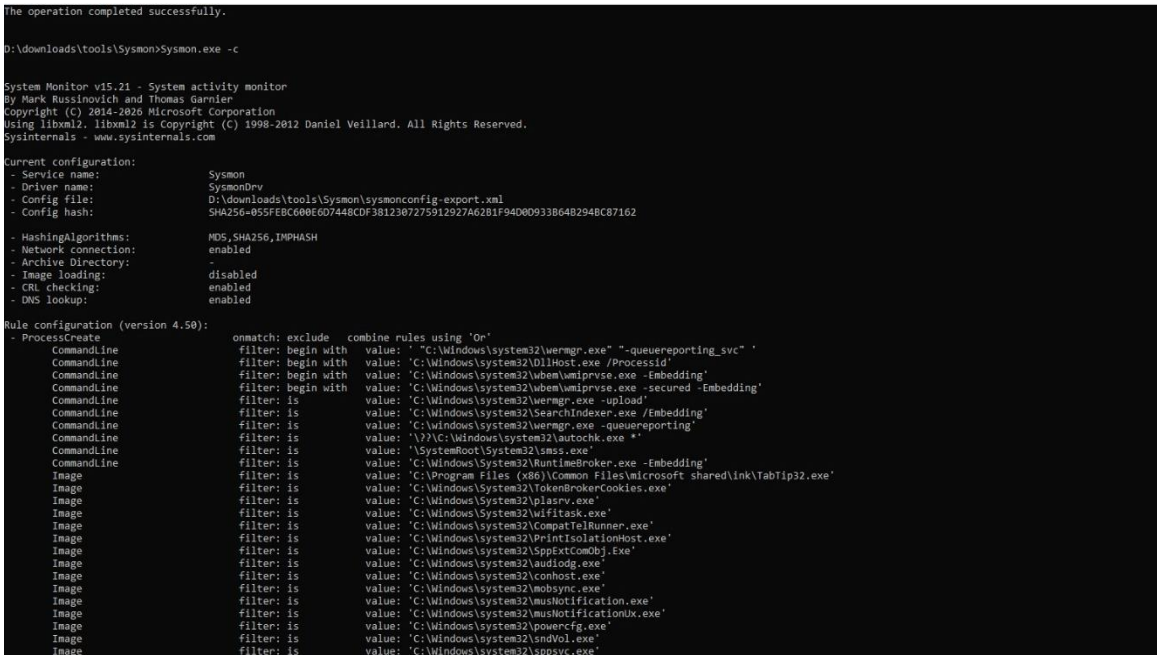
sysmon.exe -c sysmonconfig.xml

Updates the current monitoring rules without reinstalling Sysmon.

3.2. View Current Configuration

sysmon.exe -c

Displays the active Sysmon configuration.



3.3. Check Sysmon Version

sysmon.exe -?

or

sysmon.exe --version

Shows the installed version and available command-line options.

3.4. Uninstall Without Confirmation

sysmon.exe -u force

Removes Sysmon immediately without asking for confirmation.

3.5. Help Command

sysmon -h

-h or --help displays the Sysmon help menu.

```
D:\downloads\tools\Sysmon>Sysmon --help

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Usage:
Install:                Sysmon.exe -i [<configfile>]
Update configuration:   Sysmon.exe -c [<configfile>]
Install event manifest: Sysmon.exe -m
Print schema:           Sysmon.exe -s
Uninstall:              Sysmon.exe -u [force]
  -c    Update configuration of an installed Sysmon driver or dump the
        current configuration if no other argument is provided. Optionally
        take a configuration file.
  -i    Install service and driver. Optionally take a configuration file.
  -m    Install the event manifest (done on service install as well)).
  -s    Print configuration schema definition of the specified version.
        Specify 'all' to dump all schema versions (default is latest)).
  -u    Uninstall service and driver. Adding force causes uninstall to proceed
        even when some components are not installed.

The service logs events immediately and the driver installs as a boot-start driver to capture activity from early in the boot that the service will write to the event log when it starts.

On Vista and higher, events are stored in "Applications and Services Logs/Microsoft/Windows/Sysmon/Operational". On older systems, events are written to the System event log.

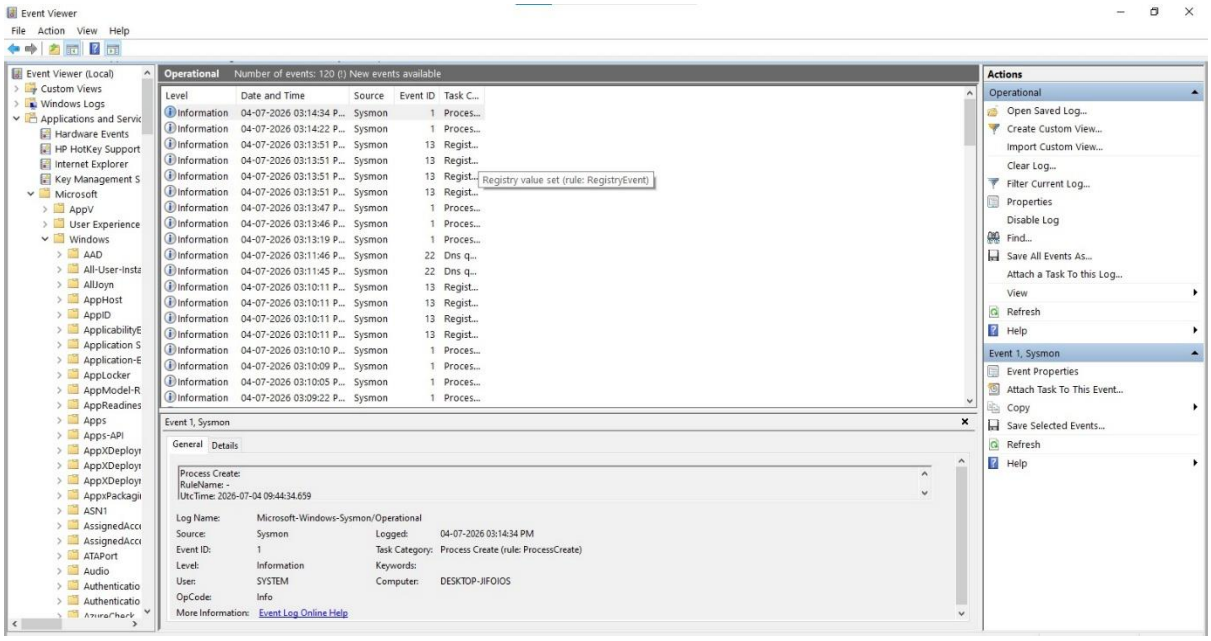
Use the '-? config' command for configuration file documentation. More examples are available on the Sysinternals website.

Specify -accepteula to automatically accept the EULA on installation, otherwise you will be interactively prompted to accept it.

Neither install nor uninstall requires a reboot.
```

Example:

Monitoring system activities in real time.



4. Introduction to theharvester tool

theHarvester is an Open Source Intelligence (OSINT) tool used to collect publicly available information such as email addresses, subdomains, hostnames, IP addresses, and employee names about a target domain. It is commonly used during the reconnaissance phase of cybersecurity assessments.

5.Setting up for theHarvester

5.1. Update the System

sudo apt update

Updates the package list from the repositories.

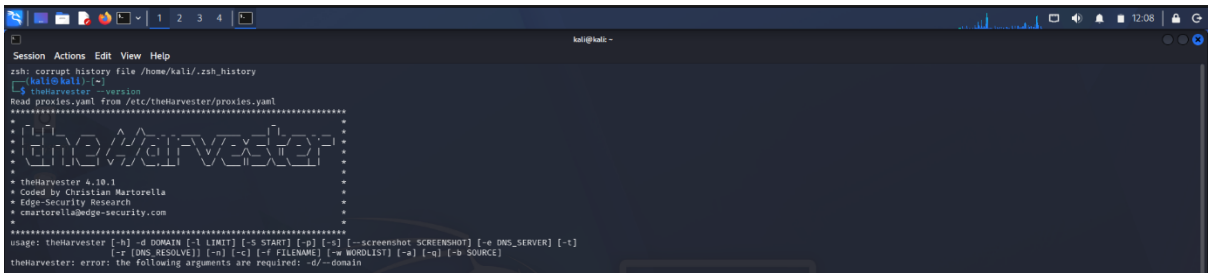
5.2. Install theHarvester

sudo apt install theharvester

Installs theHarvester from the distribution's package repository.

5.3. Verify Installation

Displays the installed version of theHarvester.



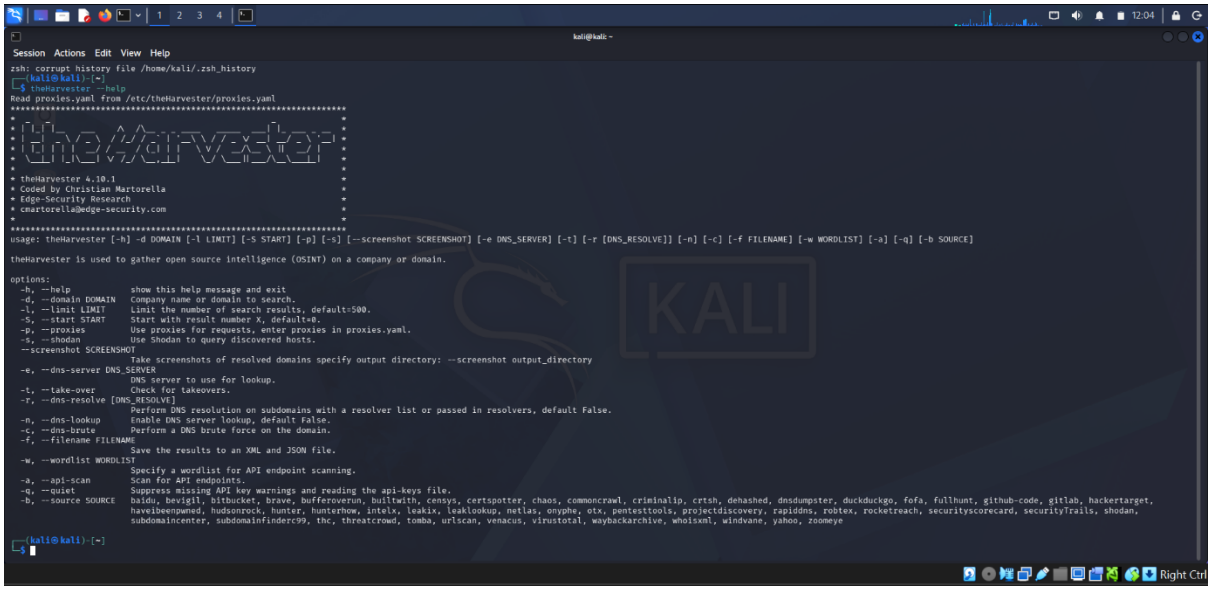
5.4. Display Help

theHarvester -h

or

theHarvester –help

Shows all available options, syntax, and examples.



Basic Syntax

theHarvester -d <domain> -b <source>



- -d → Target domain
- -b → Data source to search

  Example : theHarvester -d microsoft.com -b all -l 300 -q

6. Introduction to Netdiscover

Netdiscover is a network reconnaissance tool used to discover live hosts on a local network by sending ARP (Address Resolution Protocol) requests. It is commonly used to identify the IP and MAC addresses of devices connected to a LAN.

7. Setting up netdiscover

7.1. Update the package list

sudo apt update

Updates the package repository information.

7.2. Install Netdiscover

sudo apt install netdiscover

Installs Netdiscover on Ubuntu, Debian, or Kali Linux.

7.3. Verify Installation

netdiscover --version

Displays the help menu and confirms that Netdiscover is installed.



Basic Syntax

netdiscover [OPTIONS]

8. Commands for netdiscover

8.1. Display Help

netdiscover -h

Shows all available options and command syntax.



8.2. Scan the Default Network Interface

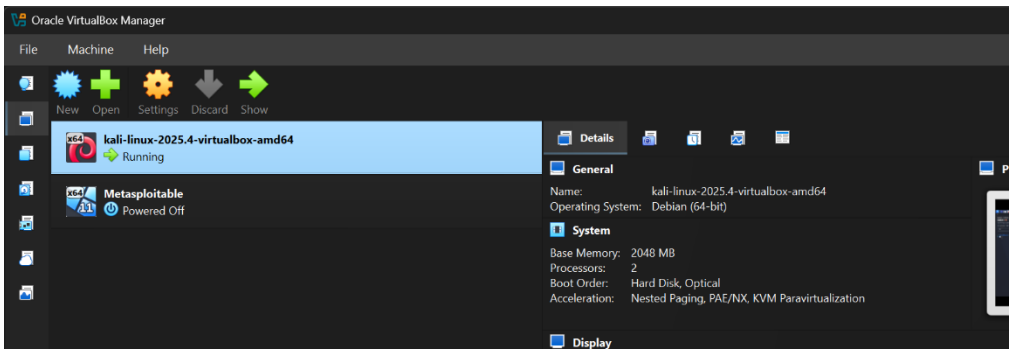
sudo netdiscover

Starts scanning the local network using the default network interface.

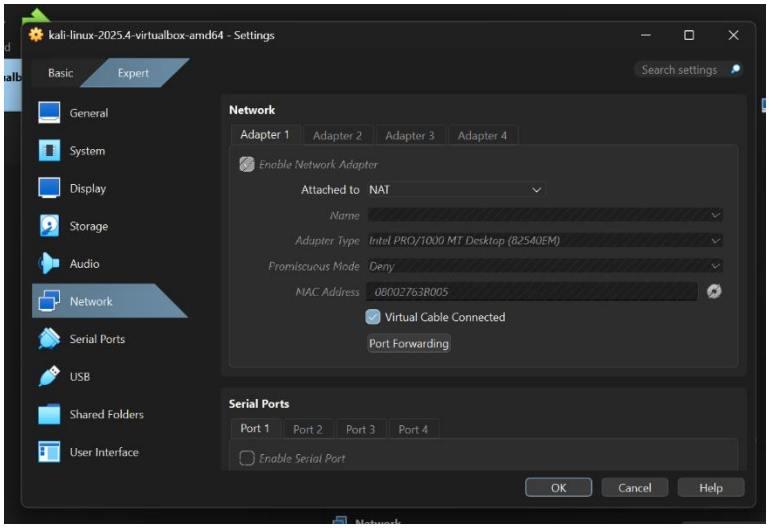
8.3. Scan a Specific Network Range

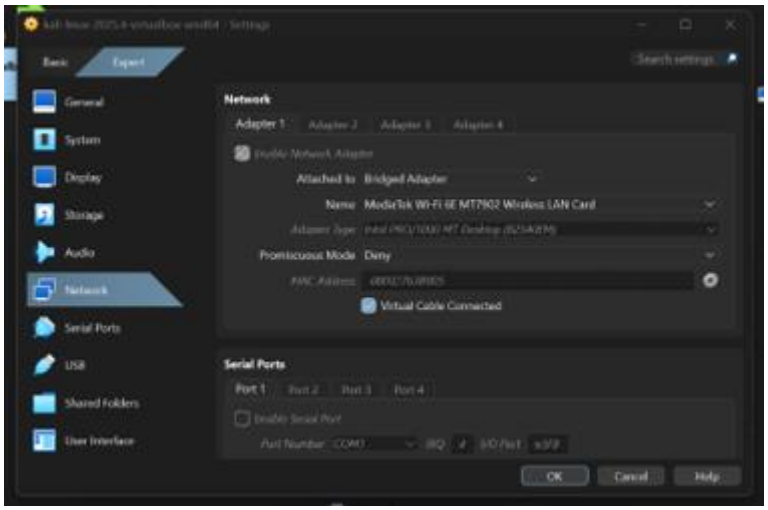
Setting up for netdiscover in the virtualbox

1.Open the settings in the virtual box



2. Enter into network and change NAT to Bridged Adapter in the attached to dropdown option.





sudo netdiscover -r 192.168.1.0/24

- -r → Specifies the IP range to scan.
- 192.168.1.0/24 → Scans all devices in that subnet.



8.4. Scan Using a Specific Network Interface

sudo netdiscover -i eth0

- -i → Specifies the network interface.
- eth0 → Uses the eth0 interface for scanning.



8.5. Scan a Specific Interface and Range

sudo netdiscover -i wlan0 -r 192.168.0.0/24

Scans the specified subnet using the wlan0 wireless interface.

8.6. Passive Mode Scan

sudo netdiscover -p

- -p → Enables passive mode.
- Listens for ARP traffic without actively sending ARP requests.



8.7. Ignore Home Network

sudo netdiscover -S

-S → Skips scanning the local/home network and focuses on other detected networks

9. Introduction to WHOIS

WHOIS is a command-line tool used to retrieve registration information about a domain name or IP address. It provides details such as the registrar, registration dates, nameservers, and registration status.

10. Setup for Whois

10.1. Update the package list

sudo apt update

Updates the package repository information.

10.2. Install WHOIS

sudo apt install whois

Installs the WHOIS command-line tool.

10.3. Verify Installation

whois –version



Displays the installed version of the WHOIS tool (if supported).

Basic Syntax

whois <domain_or_ip>

Example:

whois google.com



11. Whois Commands

11.1. Display Help

whois –help

Displays available options and command usage.



11.2. Look Up a Domain

whois example.com

Retrieves registration details for the domain example.com.

11.3. Look Up an IP Address

whois 8.8.8.8

Displays registration information for the specified IP address.



11.4. Query a Specific WHOIS Server

whois -h whois.verisign-grs.com example.com

- -h → Specifies the WHOIS server to query.
- Useful when you want to query a particular registry.

